EXP 4

Aim

To simulate microwave mixer frequency conversion using an RF signal of 10 GHz and an LO signal of 9.5 GHz, identify the intermediate frequency (IF) and sum-frequency components, and calculate the conversion loss for a 50 Ω IF load.

Objectives of this Experiment:

1.To develop simulation models representing the operating characteristics of selected microwave semiconductor devices.
2.To analyze the electrical characteristics and frequency-dependent behavior of Gunn, Schottky or PIN diode-based microwave structures.
3.To simulate frequency conversion in a microwave mixer and identify the generated frequency components.
4.To evaluate conversion characteristics and identify suitable operating conditions for microwave applications**.**

Software required:

MATLAB

Program:

```
clc;
clear;
close all;

% Microwave Mixer Frequency Conversion and Conversion Loss

% Given parameters
fRF = 10e9;              % RF frequency = 10 GHz
fLO = 9.5e9;             % LO frequency = 9.5 GHz
PRF_dBm = -10;           % RF input power
PLO_dBm = 7;             % LO input power
R = 50;                  % IF load resistance
```

```
grid on;
xlabel('Frequency (GHz)');
ylabel('Power (dBm)');
title('Microwave Mixer Output Frequency Spectrum');
xticks([0.5 9.5 10 19.5]);
xlim([0 21]);

% Part A: Frequency deviation
fIF_sim = 0.5e9;
fSUM_sim = 19.5e9;

IF_deviation = abs(fIF_sim - fIF)/fIF * 100;
SUM_deviation = abs(fSUM_sim - fSUM)/fSUM * 100;
```

```
% Theoretical frequency components
fIF = abs(fRF - fLO);
fSUM = fRF + fLO;

fprintf('MICROWAVE MIXER SIMULATION\n');
fprintf('--------------------------\n');
fprintf('RF Frequency = %.2f GHz\n',
fRF/1e9);
fprintf('LO Frequency = %.2f GHz\n',
fLO/1e9);
fprintf('Theoretical IF Frequency = %.2f
GHz\n', fIF/1e9);
fprintf('Theoretical Sum Frequency =
%.2f GHz\n', fSUM/1e9);

% Simulated spectrum: idealized
frequency-conversion model
% Assumed conversion loss = 8 dB, within
expected 6-10 dB range
CL_dB = 8;
PIF_dBm = PRF_dBm - CL_dB;

% Relative spectrum amplitudes in dBm
f_components = [fIF fRF fLO fSUM];
P_components = [PIF_dBm PRF_dBm PLO_dBm
...
    PRF_dBm - CL_dB];

% Plot output spectrum
figure;
stem(f_components/1e9, P_components,
'filled', ...
    'LineWidth', 1.5);
```

```
fprintf('\nPART A: FREQUENCY
CONVERSION\n');
fprintf('IF Frequency = %.2f GHz\n',
fIF_sim/1e9);
fprintf('Sum Frequency = %.2f GHz\n',
fSUM_sim/1e9);
fprintf('IF Frequency Deviation = %.2f
%%\n', IF_deviation);
fprintf('Sum Frequency Deviation = %.2f
%%\n', SUM_deviation);

% Part B: Conversion loss
fprintf('\nPART B: CONVERSION LOSS\n');
fprintf('RF Input Power = %.2f dBm\n',
PRF_dBm);
fprintf('LO Input Power = %.2f dBm\n',
PLO_dBm);
fprintf('Assumed IF Output Power = %.2f
dBm\n', PIF_dBm);
fprintf('Conversion Loss = %.2f dB\n',
CL_dB);

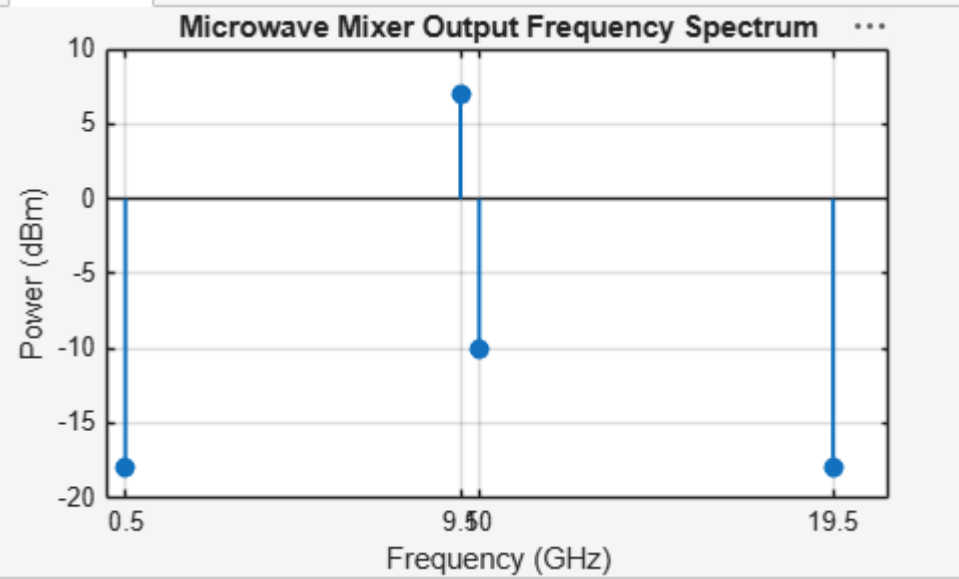
if CL_dB >= 6 && CL_dB <= 10
    fprintf('Conversion loss is within
the expected range.\n');
else
    fprintf('Conversion loss is outside
the expected range.\n');
end

% Power conversion to watts
PIF_W = 1e-3 * 10^(PIF_dBm/10);
VIF_rms = sqrt(PIF_W * R);

fprintf('IF Output Power = %.4e W\n',
PIF_W);
fprintf('IF Load RMS Voltage = %.4f
V\n', VIF_rms);
```

Output:

```
Conversion Loss = 8.00 dB
Conversion loss is within the expected range.
IF Output Power = 1.5849e-05 W
IF Load RMS Voltage = 0.0282 V
>>
```


Microwave Mixer Output Frequency Spectrum
10
5
0
-5
-10
-15
-20
Power (dBm)
0.5
9.5
10
19.5
Frequency (GHz)